Junhao Zhang

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EDUCATION

University of Illinois Urbana-Champaign **Expected Graduation: May 2029**
Bachelor of Science in Mathematics & Computer Science Cumulative GPA: 3.99/4.00
Bachelor of Arts in Spanish

SKILLS

Technical Skills: C/C++, Rust, TypeScript, React, Node.js, Java, Python, Git, LaTeX
Languages: English (Native), Chinese (Native), Spanish (Advanced Conversational Proficiency)

RELEVANT COURSEWORK

CS 225 (Data Structures in C++), **STATS 400** (Stats & Probability), **CS 128H** (Intro to Rust), **CS 357** (Numerical Methods)
CS 233 (Computer Architecture), **CS 374** (Intro to Algorithms and Models of Computation)

EMPLOYMENT AND RESEARCH

CS128H | Course Assistant | Urbana, IL **Jan 2026 – Present**
- Host weekly office hours and provide one-on-one tutoring to clarify course concepts
- Design lecture presentations, and homework assignments for 60+ students in an introductory Rust course
- Collaborate with instructors to refine course content and improve learning outcomes

Punyasena Paleoecology Lab, SIB | Student Researcher | Urbana, IL **Jan 2026 – Present**
- Develop two-stage computer vision pipeline using PyTorch and SAM2 (Segment Anything Model) to segment pollen grains
- Design heuristic-driven prompting strategies resulting in accurate segmentation of all test data (30 samples)
- Present results in research poster at the Undergraduate Research Symposium with 750+ participants

Cloudflare | Software Engineer Intern | Austin, TX **May 2026 – Aug 2026**
- Built and shipped a full-stack debugging tool for the WARP VPN client in React and Rust/WASM
- Refactored a prototype into production-quality code, adding agentic RAG-based lookup for historical issue resolutions
- Worked with a PM and 7-person engineering team to understand existing workflows guide feature design

PROJECTS

Europa Lang | github.com/cursorweb/europa-lang | Rochester, MN **Sept 2021 - Present**
- Built a custom language in Rust with a hand-written lexer, recursive-descent parser, closures, and first-class functions
- Developed a REPL, standard library, and VS Code extension, coordinated 4 contributors across tooling and runtime features

RusTX | github.com/cursorweb/raytracing-in-rust | Urbana, IL **Nov 2025 – Dec 2025**
- Led a team of 3 to design and implement a ray tracer in Rust as a CS128 Honors final project
- Applied Rust's concurrency features to create a multi-threaded renderer reducing render time from 10 minutes to 2 minutes

CrabPen | github.com/CursorwebGames/2DThreePen | Rochester, MN **May 2026 - Present**
- Build Rust/WASM puzzle generator using Dancing Links to create unique-solution Queens puzzles in-browser
- Used benchmarks to apply targeted optimizations, cutting generation time for 14x14 boards from 13.2s to 200ms

ACTIVITIES

University of Illinois Symphony Orchestra | Violinist | Urbana, IL **Aug 2025 – Present**
- Perform in an Orchestra of music performance majors as well as highly talented non-majors
- Hold concerts 3 times per semester, with a variety of repertoire including works by Brahms, Shostakovich, Mahler, and Beethoven

Volunteer Income Tax Assistant (VITA) | Volunteer | Urbana, IL **Jan 2026 – Present**
- Contribute to the completion of 300+ tax returns in the community by helping verify eligibility for VITA services
- Provide Spanish-English translation, supporting community members via phone

HONORS AND AWARDS

Dean's List **All Semesters**

Minnesota Bilingual Seal (Spanish and Chinese) - Platinum Level **2024, 2025**
- Demonstrated ACTFL advanced low proficiency in Speaking, Writing, Reading, and Listening in Chinese and Spanish